

Complementary Slackness

Opt Condition for LP

$$\begin{aligned} \min c^T x \\ \text{s.t. } Ax \geq b \\ x \geq 0 \end{aligned}$$

Then dual is:

$$\begin{aligned} \max b^T y \\ \text{s.t. } A^T y \leq c \\ y \geq 0 \end{aligned}$$

Corollary weak duality is that if $c^T x = b^T y$ then they are optimal

Thm If primal(dual) has finite optimal $x^*(y^*)$ then so does the dual (primal) $y^*(x^*)$

$A : p \times q$ where $p \neq$ constraints but $A^T : q \times p$ for dual and p is the # variables

y_i is the dual variable which corresponds to the i th constraint in the primal

Slack in primal (excess): $e = Ax - b \geq 0$ if x is feasible (e has p variables)

Slack in dual: $s = c - A^T y \geq 0$ if y is feasible (s has q variables)

Thm Let $\bar{x}(\bar{y})$ be a feasible solution of the primal (dual) then $\bar{x}, \bar{y}$ optimal iff:

$$\forall i \in \{1, \dots, p\} : e_i \bar{y}_i = 0, \forall j \in \{1, \dots, q\} : s_j \bar{x}_j = 0$$

EX

$$\begin{aligned} \min x_1 + 2x_2 \\ 3x_1 + 4x_2 \geq 5 \end{aligned}$$

and dual is

$$\begin{aligned} \max 5y \\ 3y \leq 1 \\ 4y \leq 2 \end{aligned}$$

So the equalities are

$$(3x_1 + 4x_2 - 5)y = 0$$

$$(1 - 3y)x_1 = 0$$

$$(2 - 4y)x_2 = 0$$

Proof

$$e^T \bar{y} = \sum_i e_i \bar{y}_i, s^T \bar{x} = \sum_j s_j \bar{x}_j$$

→ If $\bar{x}, \bar{y}$ optimal then by strong duality, $c^T \bar{x} - (\bar{y})^T A \bar{x} = 0$

$$= (c^T - (\bar{y})^T A) \bar{x}$$

$= s^T \bar{x} = 0$ since all variables are positive then the individual sum of $s_j \bar{x}_j$ are all 0

$$\text{Similarly: } \bar{y}^T A \bar{x} - b^T \bar{y} = 0$$

$= e^T \bar{y} = 0$ since all variables are positive then the individual sum of $s_j \bar{x}_j$ are all 0

$$\text{Now } \leftarrow e_i \bar{y}_i = 0 \forall i, s_j \bar{x}_j = 0 \forall j$$

It is the exact same idea.... start with weak duality and show that they are equal by substituting in our slack definition so now by the corollary it is optimal

This means $e_i > 0 \rightarrow \bar{y}_i = 0$ similarly $s_j > 0 \rightarrow \bar{x}_j = 0$

EX

$$\min 3x_1 + 4x_2$$

$$5x_1 + 6x_2 = 7$$

in sym form:

$$\min 3x_1 + 4x_2$$

$$5x_1 + 6x_2 \geq 7$$

$$-5x_1 - 6x_2 \geq -7$$

in dual

$$\max 7y_1 - 7y_2$$

$$5y_1 - 5y_2 \leq 3$$

$$6y_1 - 6y_2 \leq 4$$

Suppose $z = y_1 - y_2$, then:

$$\max 7z$$

$$5z \leq 3$$

$$6z \leq 4$$

$$\text{With } z \leq 0$$

In other words we can take the dual of a standard form by just unrestricting the z